Q5. Histogram for EDA

Original Aim: Perform EDA on the Iris dataset by plotting a histogram of PetalLengthCm and a scatter plot of SepalLengthCm against PetalLengthCm. Add suitable titles, axis labels and different colours for Species in the scatter plot. Briefly state what the plots reveal about the distribution and relationship between the attributes.

Aim

Plot a PetalLengthCm histogram and a species-coloured scatter plot of SepalLengthCm against PetalLengthCm, with labels and interpretation.

Requirements

Python, Pandas, scikit-learn and Matplotlib. A notebook or graphical IDE displays plots; each figure is also saved to a PNG.

Essential theory

A histogram groups a numerical variable into bins and displays frequency, revealing shape, gaps and concentration. A scatter plot represents paired measurements as points and helps examine association and groups. Species colours allow comparison of known classes. In the phrase "sepal length against petal length," this solution places petal length on x and sepal length on y; both axes are explicitly labelled.

Step-by-step procedure

1. Load Iris and create one figure with two axes.
2. Plot the PetalLengthCm histogram using 12 bins.
3. Group by species and plot its paired measurements in a distinct colour.
4. Add titles, centimetre axis labels, frequency label and species legend.
5. Tighten the layout, save/display the figure and inspect the observed species ranges.

Complete Python program

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
features = [
    "SepalLengthCm", "SepalWidthCm",
    "PetalLengthCm", "PetalWidthCm"
]
species = ["Iris-" + name for name in iris.target_names]
df = pd.DataFrame(iris.data, columns=features)
df["Species"] = pd.Categorical.from_codes(
    iris.target, categories=species
)

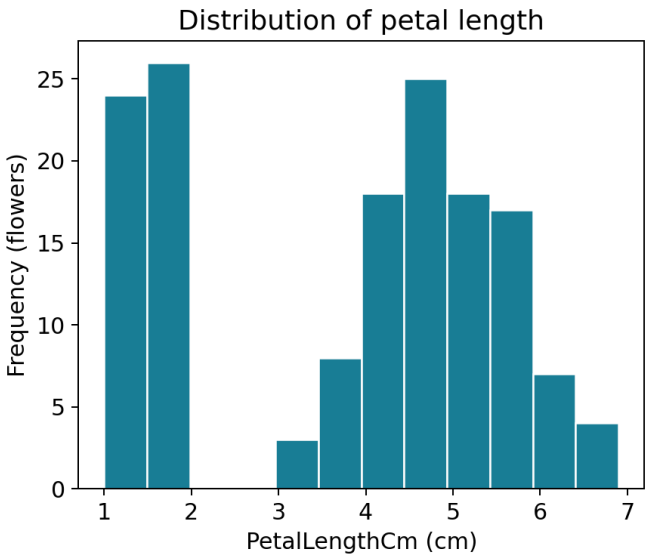
plt.rcParams.update({"font.size": 12})
fig, axes = plt.subplots(1, 2, figsize=(10, 4.4))
axes[0].hist(df["PetalLengthCm"], bins=12,
             color="#187d95", edgecolor="white")
axes[0].set_title("Distribution of petal length")
axes[0].set_xlabel("PetalLengthCm (cm)")
axes[0].set_ylabel("Frequency (flowers)")
colors = ["#147d92", "#dc8b26", "#7357a6"]
for label, color in zip(species, colors):
    group = df[df["Species"] == label]
    axes[1].scatter(group["PetalLengthCm"], group["SepalLengthCm"],
                    label=label, color=color, alpha=0.8, s=28)
axes[1].set_title("Sepal length against petal length")
axes[1].set_xlabel("PetalLengthCm (cm)")
axes[1].set_ylabel("SepalLengthCm (cm)")
axes[1].legend(fontsize=11)
fig.tight_layout()
fig.savefig("q05_eda.png", dpi=180)
plt.show()
print("Saved q05_eda.png")
print("PETAL LENGTH RANGES BY SPECIES")
print(df.groupby("Species", observed=True)["PetalLengthCm"]
        .agg(["min", "max"]).to_string())
```

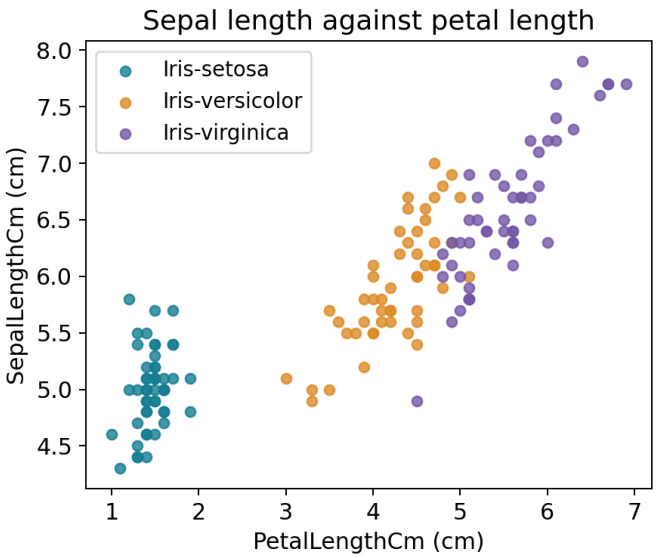
Code explanation

- `plt.subplots(1, 2)` creates two side-by-side axes in one figure.
- `hist(..., bins=12)` counts observations within 12 intervals; bin choice affects the appearance, not the underlying measurements.
- Boolean selection `df[df["Species"] == label]` isolates one species. The loop plots all three known classes with a readable legend.
- `alpha` reduces overplotting and `s` controls marker area. `tight_layout()` reduces clipping.
- `savefig` runs before `show`, retaining the figure even in environments that close it after display. The numerical range table supports the visual interpretation.

Verified output and figure

```
Saved q05_eda.png
PETAL LENGTH RANGES BY SPECIES
                 min  max
Species
Iris-setosa      1.0  1.9
Iris-versicolor  3.0  5.1
Iris-virginica   4.5  6.9
```





Observation

Setosa has short petals (1.0-1.9 cm) separated from the larger-petal groups. Versicolor ranges from 3.0 to 5.1 cm and virginica from 4.5 to 6.9 cm, so their ranges overlap. Across the pooled data, sepal length tends to increase with petal length, while species groups contribute to that pattern. A pooled trend need not have the same strength within each species. The histogram is not a single symmetric bell-shaped group.

Result

The requested histogram and species-coloured scatter plot were generated, labelled, saved and interpreted using actual dataset values.

Common errors and troubleshooting

- Blank saved figure: save the current figure before show/close, as in the code.
- Plot not displayed in a noninteractive terminal: inspect the saved PNG or use a notebook/graphical backend.
- Missing legend: give each species series a label before calling legend().

Viva questions and answers

- What does a histogram's y-axis represent here? The number of flowers in each petal-length bin.
- How does a scatter plot differ? It shows paired values of two numerical variables for each observation.
- Why colour points by species? It reveals grouping and overlap hidden in a pooled plot.
- Does an upward scatter pattern prove causation? No; it shows association, potentially influenced by groups.